

Closure Properties of Regular Languages

Concatenation Operation

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Objectives

This tutorial examines the concatenation operation on regular languages and demonstrates the closure property of regular languages under concatenation. Students will develop understanding through progressive exercises ranging from basic string concatenation to formal language operations.

1 String Concatenation

Question 1. Perform the concatenation operation on the following pairs of strings over the alphabet $\Sigma = \{a, b, 0, 1\}$:

- (a) $w_1 = \text{"ab"}, w_2 = \text{"cd"}$
- (b) $w_1 = \text{"0"}, w_2 = \text{"11"}$
- (c) $w_1 = \text{"101"}, w_2 = \text{"0"}$

2 Language Concatenation: Finite Languages

Question 2. Consider the languages $L_1 = \{a, b\}$ and $L_2 = \{0, 1\}$ over the alphabet $\Sigma = \{a, b, 0, 1\}$.

- (a) Enumerate all strings in the concatenation L_1L_2 .
- (b) Determine the cardinality of L_1L_2 .

Question 3. Let $L_1 = \{0, 00\}$ and $L_2 = \{1, 11\}$ be languages over $\Sigma = \{0, 1\}$. Complete the following table to systematically determine all elements of L_1L_2 :

String from L_1	String from L_2	Concatenated String
0	1	
0	11	
00	1	
00	11	

3 Conceptual Understanding

Question 4. Consider two deterministic finite automata M_1 and M_2 that recognize regular languages L_1 and L_2 respectively.

- (a) Describe the role of machine M_1 in the concatenation operation.
- (b) Describe the role of machine M_2 in the concatenation operation.
- (c) Specify the order of execution for machines M_1 and M_2 when recognizing strings in L_1L_2 .

4 Language Concatenation: Infinite Languages

Question 5. Define the following infinite regular languages: $L_1 = \{a^n \mid n \geq 1\}$ and $L_2 = \{b^m \mid m \geq 1\}$.

- (a) Provide three distinct elements of L_1L_2 .
- (b) Give a precise characterization of L_1L_2 in natural language.

Question 6. Consider the languages $L_1 = \{\varepsilon, a\}$ and $L_2 = \{b, bb\}$ where ε denotes the empty string.

- (a) Determine all elements of L_1L_2 .
- (b) Explain the behavior of the empty string ε under the concatenation operation.

Question 7. Let $L_1 = \{0^n \mid n \geq 1\}$ and $L_2 = \{1\}$ be regular languages over $\Sigma = \{0, 1\}$.

- (a) Provide three representative elements of L_1L_2 .
- (b) Express L_1L_2 using formal set-builder notation.

5 Non-Commutativity of Concatenation

Question 8. Consider the languages $L_1 = \{a, aa\}$ and $L_2 = \{b, ab\}$ over $\Sigma = \{a, b\}$.

- (a) Compute L_1L_2 .
- (b) Compute L_2L_1 .
- (c) Determine whether $L_1L_2 = L_2L_1$. What property of the concatenation operation does this demonstrate?